1 Planning (15 marks)

Defining the problem (3 marks)

P1 x is the independent variable, B is the dependent variable or vary x and measure B [1]

P2 Keep the number of turns on the coil/radius of the coil constant [1]
Do not accept same coil – 'coil' is not a variable

P3 Keep the current (in the coil) constant [1]

Methods of data collection (5 marks)

M1 Diagram showing coil and Hall probe with a means of read out appropriately positioned along axis [1]

M2 Coil connected to a power supply [1]

M3 Measure x with a ruler [1]

M4 Hall probe at right angles to direction of magnetic field or gives maximum output for each reading [1]

M5 Method to determine axis of coil or to find $x = 0$ [1]

Method of analysis (2 marks)

A1 Plot a graph of ln B against x [1]

A2 Relationship valid if a straight line is produced (ignore reference to y-intercept) [1]

Safety considerations (1 mark)

S1 Precaution linked to (large) current in coil/heating, e.g. switch off when not in use to avoid overheating coil; do not touch because it is hot [1]

Additional detail (4 marks)

D 1/2/3/4 Relevant points might include [4]

1. Method to create a large magnetic field, e.g. use large current or large number of turns.
2. Reasoned method to keep current constant.
3. Reasoned method to keep Hall probe in same orientation (e.g. use of set square, fix to rule, optical bench or equivalent).
4. B is proportional to voltage across Hall probe/calibrate Hall probe (in a known magnetic field).
5. Repeat experiment with Hall probe reversed or equivalent.
6. Identifies logarithmic equation i.e. $\ln B = -p\,x + \ln B_0$
7. Avoid external magnetic fields.
8. Method to keep Hall probe along axis.

Do not allow vague computer methods.

[Total: 15]

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